

Research Proposal and Personal Fit.

Sid Diamond

Contents

1	Document Abstract	2
1.1	Document Contents	2
1.2	Open Abstract	2
2	Theory of Impact	3
2.1	Importance	3
2.2	Neglect	4
2.3	Tractability	6
3	Personal Motivation	7
3.1	Personal Fit	7
3.2	Personal Uncertainty	7
3.3	Skills required	8
4	MATS feedback	9

1 Document Abstract

1.1 Document Contents

Contents: This document contains a research proposal birthed out of assessing where I fit in high impact work. It contains an abstract for a research project, a case for the proposal being high impact and a discussion of my own personal fit doing a project such as the one proposed. It also concludes by containing my MATS application feedback.

Process: In attempting to follow the exercises in 80,000 hours' [How to find the right career for you](#), my initial list of potential careers mutated into a brain dump of venn diagrams and arrows pointing to vague concepts, roles and technical jargon. I figured the best thing to do was instead create a made-up job role that, to differing extents, included the things within the brain dump. Long story short, I accidentally created a pretty open-ended technical AI safety research proposal.

Result: Currently, I'm interested in measuring how differences in some set of quantities, S , relating to a model's character, cause emergent complexity in multi agent¹ networks and how this can inform on safer model design. This is best illustrated through a research proposal. This proposal is summed up by the so-called *open abstract* (as it is more open-ended compared to a finished research paper abstract) detailed below.

1.2 Open Abstract

We monitor how individual agents'^{1,1} sense of self-identity evolve in a multi-agent, sandboxed, alignment experiment. We extend [Douglas et al.](#)'s modification to the blackmail (or otherwise) experiment by adding multiple agents with different self-conceptions and allow them to interact. We study:

- I) What emergent properties the experimental setup exhibits and how robust these are to setup perturbations. Example properties for a given initial state include,
 - a. What the ultimate outcome of the task set to the agents is.
 - b. What equilibrium of self-identities, or otherwise, in the system are reached.
- II) What kinds of inter-agent model collaboration or sabotage occur, investigated using a so-called swiss-cheese approach of layering multiple flawed mechanistic interpretability techniques, most likely the combination of CoT (Chain of Thought) and J-space monitoring.
- III) Extensions could include: What happens if there is a human in the loop? Different initial conditions. Robustness to different sizes of population, model types and model scaffolds.

¹A technical point here is that 'agent' is used to refer to the individual unit of AI system, however we acknowledge a different kind of scaffold or no scaffold at all may be used in practice.

2 Theory of Impact

2.1 Importance

2.1.1 Low Hanging Fruit in Alignment

Model character can alter actions a model takes to pursue a goal. “*Varying a model’s identity boundary can sometimes shift its behaviour as much as varying its goals.*” [1]. This has been shown empirically in an adapted version of the model blackmail experiment [1, 2].

The bet of organisations like [Forethought](#) and [Differential Labs](#) is that within alignment research, there exist low-hanging, highly impactful, fruit [3]. What is meant by this is that whilst alignment policy that surrounds issues humans commonly disagree on is very difficult, alignment choices where humans largely agree is, in comparison, far easier. For example, broadly speaking, humans would be better served by developing AI systems that make conflict resolution easier rather than systems that make killing people easier. Yet there appears to be substantially more research into military AI capabilities than into conflict-resolution or macrostrategy AI capabilities [3]. If AI systems continue to become more capable and more autonomous, this imbalance becomes increasingly consequential. With that in mind, it is not unreasonable to argue that reducing the capabilities gap between frontier military AI systems and conflict-resolution AI systems would be valuable, even if doing so involves capabilities research that carries some risk of creating unforeseen harms [4]. Conducting such research in controlled sandbox environments could also help surface and address some of those risks before deployment. Given the difficulty of slowing progress in military AI [5], there is therefore a case for improving the capabilities of conflict-resolution AI systems in parallel, potentially buying valuable time for work on alignment, governance and interpretability. Compared to some alignment challenges, performing at least some kind of *AI for geopolitical macro-strategy research*, is low hanging fruit. This example, in fact, is the bet of [Differential Labs](#) [3].

So what is the low hanging fruit the *open abstract* is hoping to pluck?

2.1.2 AI Character Choices

As established, AI character generates tangible impact on goal outcome [1]. We currently exist at a time in AI capabilities where we can not only study these characters, but engineer them to be closer to our liking based on the results of safe (if done carefully), sandboxed experiments. If we can engineer model character in ways that lead to generally safer outcomes in ways most people don’t have a problem with, we probably should. This would resemble plucking low-hanging fruit.

In the case of single-identity scaffolds, such as a distinct agent, whilst this is still not worked on enough (so much so it is one of Coefficient Giving’s [Project Tailwind](#) initiatives), model character is an area of active research. The frontier of this research is probably Anthropic’s Claude’s Constitution work [6]. However, as is true in humans and animals, the result of multi-character interactions is an emergent phenomena that cannot be understood simply from the list of characters interacting. As put by network scientist Albert-László Barabási, “*Society is not a phone book. What makes society work is really the interactions among us*” [7]. Multi-agent^{1.1} character interaction studies using the same top-down logic prominent in more established areas of complex systems research is needed [8]. The bet in this proposal is:

- I) The type of emergent complexity seen in the interaction of scaffolded models working together to accomplish a task is causally related to the character’s the model’s in the system have.
- II) Editing these model characters within sandboxed alignment experiments can produce safer outcomes with little negative impact on performance in control tasks.

- III) These sandbox experiments could either directly inform real-world decisions or motivate larger studies that do so, ultimately helping to make interactions between AI systems safer.

So, if the bet in this proposal is correct, it represents low hanging fruit in terms of AI safety. This begs the question, if the bet in this proposal is correct, why aren't more people making it?

2.2 Neglect

The short answer is, like many areas in AI safety, both multi-agent interaction and model character are neglected. More people should be working on both of these areas and this results in a neglected area intersection. We now put forward a brief case for each of these research areas being neglected.

In the case of multi-agent interaction, the [Cooperative AI Foundation](#), whose mission is to support research into the cooperative intelligence of advanced AI, launched a \$10m multi-agent safety funding call in 2026 alongside Schmidt Sciences, Google DeepMind and ARIA [9]. In motivating the call, they argue that *“more research is therefore urgently needed to understand safety and risk through a system-level, multi-agent lens.”* Their research agenda specifically calls for work on emergent collective dynamics, including how collective capabilities arise and how dangerous population-level properties can be detected. They even outline the desire for the research they intend to fund to be [important, tractable and neglected](#).

The case for model character being neglected can be seen through the aforementioned project tailwind call for new *model character evaluator* organisations. Moreover, this case can also be seen on the Forethought website, with the top suggestion under their website subheading *“act”* being the following:

“[AI character evaluation](#). Start an independent org to evaluate and stress-test AI character traits (epistemic integrity, prosociality, appropriate refusals), hold developers accountable against their own model specs / constitutions, and suggest and incentivise improvements to the specs.” [10].

As a result, the combination of the two: multi-agent dynamics and character evaluation, is of compounded neglect and importance. Moreover, there exists, to our best knowledge, not nearly enough openly-visible research in this area. AI safety needs more researchers plucking this low-hanging fruit.

With that said, there do exist some research organisations working at the intersection of multi-agent and model character research. For example, [ACS Research](#), the [Cooperative AI Foundation](#) and the [Complexity Science Hub](#). ACS research run a *LLM Psychology and Sociology programme* hoping to investigate similar questions to that of the *open abstract*. In fact, as we will discuss, a March 2026 paper from them, *The Artificial Self: Characterising the landscape of AI identity* (which proposes follow up research) heavily inspired this research proposal [1].

As for the cooperative AI foundation, in June 2026, they stated, *“more research is therefore urgently needed to understand safety and risk through a system-level, multi-agent lens”* [9]. Furthermore, in their 2025 technical report, *Multi-Agent Risks from Advanced AI*, Section 3.3 discusses how empirical simulations such as agent-based models (ABMs) could be used to study selection pressures in multi-agent systems. In particular, the authors argue that if *“sufficiently well-developed benchmarks for different model characteristics [are developed], [then] we can study their robustness under different kinds of selection pressure.”* [11]. Clearly, this implies the need for greater development of model characteristic benchmarks to allow researchers to benefit from the current hype around ABMs. ² This works well in conjunction with the aforementioned

²This hype is acknowledged by complexity economists such as Farmer and Axtell, who state that ABMs have

project tailwind funding call for new *model character evaluator* organisations.

Moreover, the Complexity Science Hub have an upcoming October 2026 workshop, *From Fear to Forecast: Understanding AI Swarms* [13]. Of course, we mention all of these organisations with the caveat that none of them, to our best knowledge, have done the exact research proposed in the *open abstract*. In other words, the current fruit pickers acknowledge there are too few of them to pick all of the low-hanging fruit.

So, the work proposed in *open abstract* has not been done. However, it is significantly motivated by the last sentence in the *The Artificial Self* paper’s abstract: “*We end with key recommendations: treat affordances as identity-shaping choices, pay attention to emergent consequences of individual identities at scale, and help AIs develop coherent, cooperative self-conceptions.*” [1]. Now, whilst they give recommendations, as far as we can tell, they don’t actually create an experiment to empirically *pay attention to emergent consequences of individual identities at scale* within this paper. This is what we hope to do.

Moreover, the space of possible follow up papers is large with multiple fruitful avenues to explore. We conjecture that, at present, two papers independently studying multi-agent character interactions are very likely to reach distinct and useful conclusions. This is because character interaction dynamics can be modelled in many different ways, following different computational approaches. Existing work has already begun to study the trade-off between model capability and agent count. For example, Kim et al. (2026) hold task prompts, tools and per-system compute budgets fixed while varying coordination structure and model capability, finding that the benefits of multi-agent coordination depend strongly on the capability of the underlying model and on coordination overhead [14]. Related work has examined how diversity within multi-agent systems affects the returns to adding further agents, finding strong diminishing returns from homogeneous scaling but continued gains from introducing heterogeneous agents [15].

This motivates a slightly different question: ‘*In using a fixed amount of compute to predict larger-scale dynamics, is it more informative to study a relatively small number of frontier-model agents and extrapolate their interaction dynamics to larger populations, or to simulate much larger populations using cheaper, less capable models and extrapolate towards frontier capability?*’ To our knowledge, existing work maps parts of this capability to population-size trade-off, but does not yet directly compare how each method extrapolates. This of course is even further neglected in that these experiments do not vary model character in the same way *The Artificial Self* paper extension proposes and that we hope to investigate [1].

2.2.1 Soft Neglect Factors

This section contains the soft factors that led me to this research. I conjecture they are an important reason I landed on this research proposal. This section is written more informally and contains the names of real people I had informal conversations with so please distribute appropriately.

This proposal was written at the LEAH office under the role of Impact First Fellow at Arcadia Impact. Since starting, I’ve been privileged to enjoy some of the most interesting conversations I’ve ever had. One of which was with Noah y Seigal at the [Sentient Futures](#) social after an Impact First seminar on population ethics on 15.09.26. Noah’s take was that AI safety is no longer as neglected as it was ten years ago. My understanding was that, to Noah, what is more neglected are the emergent properties that map most naturally onto the concept of self. He recommended reading the new [Anthropic Global Workspace \(J-space\)](#) paper. Whereas his

‘come of age’ [12]. This provides a further reason for now being the right time to study sandboxed multi-agent model character interactions, since these themselves are ABMs.

interest in the results were more related to model consciousness, mine are more focused on their implications as a new alignment tool. I conjecture, in line with Section 5 of the paper itself, that the J-lens could serve as a cheap, additional interpretability tool for model character evaluation, helping to offset some of the limitations of CoT monitoring [16]. I concede that I am unsure whether this conjecture is correct; nevertheless, I would argue that it is the job of a technical AI safety researcher to find this out empirically.

Other notable discussions that I imagine shaped this research direction were with Lysander Mawby on open questions in mechanistic interpretability, as well as 1-1 discussions with Xanthe Smith, Owen Cotton-Barratt, Stefan Heimersheim, Florian Lorkowski and Aditya Adiga at EAGxOxford. Taken together, my *research-taste gut* has landed on the following conclusion:

Understanding how measurable features associated with agent character shape the emergent behaviour of interacting multi-agent systems seems to me both important and neglected.

Now, this leaves two questions: What exactly are the tractable hypotheses to test here? Is this tractability accessible to me at this time?

2.3 Tractability

I posit within the *open abstract*, new, tractable results can be extracted. Some initial candidates include:

1. In a sandboxed character evaluation alignment audit, does the J-space provide cheap added value when used in conjunction with CoT monitoring?
2. What kind of distributions or scalars casually impact equilibriums in a multi-agent^{1.1} setting and how robust are these scalars and distributions to perturbations within a chosen the experimental set up?
3. How robust are these results across different initial combinations of character types/scheming propensities/ goals?

The specifics here deviate from the original intended purpose of this exercise and will be further investigated through a small project demonstrating initial results.

3 Personal Motivation

3.1 Personal Fit

To quote, Holden Karnofsky: *"When people ask me for career advice or whatever, the usual thing I'd say is: take a bunch of options that all seem competitive, and all seem like they could be the best thing, and that it's not obvious which ones are better than others from an impact perspective. And from there I would say go with personal fit, go with the energy you feel to work on them."* [17]. I think, relative to other things I could be doing in the world, being a technical, multi-disciplinary AI safety researcher suits my skillset.

Currently, my thought process is as follows: There will be more dedicated people than me, with a better natural aptitude for, say, pure mechanistic interpretability research. The same can be said for pure model welfare, complexity economics, forecasting, mathematical philosophy and probably all areas of technical AI governance. However, for certain research questions, with a precise combination of the above topics, I will be uniquely well-suited to helping solve them.

For questions within this subset, such as the important, neglected and tractable one the *open thesis* addresses, I want to optimise the impact answering them can have by working with talented researchers across the disciplines above by acting as a connecting node between researchers.

Relating this back to the 'dream job' formulation this project spun out of, I have a propensity to see my future role as someone less clued up on all nodes in a multi-disciplinary research team but able to facilitate the best communication between them. To act as a sound board for individual researchers with enough technical background and context to be more helpful than a sole AI system. To read literature reviews and look for the best people external to the project and convince them to help. To actively work on how the research agenda might need to adjust its theory of impact. To translate the results to different stakeholders.

As far as I'm aware, the role I'm describing is a senior one. Thankfully, I love being mentored and doing nitty-gritty technical research too. I think this is why a PhD in complex systems appealed to me: still being mentored, tooling up on the technical science of how things connect, working in multi-disciplinary teams.

I'd love to know who is planning to start the independent org proposed on the Forethought website detailed in the above quote: 2.2. I wonder if the *open abstract* is similar or different to ideas they have.

3.2 Personal Uncertainty

1. How one gains the appropriate research priors required to eventually be working on a project even remotely like the *open abstract*.
2. How viable is funding as a means of living and what are the expected timelines between now and being able to do this as a full-time job?
3. That a PhD length of time isn't required. Timelines aside, is the speed of upskilling in the AI safety industry a bit narcissistic on the amount of time it takes to gain expertise? Is the bet: *Learn by doing. In 3 years be ahead of where you would be if you had done a PhD then pivoted.* This I can believe.
4. To what extent is this all techno-futurist hype I'm falling for. If it is, what does that change? How does participating in the AI safety space help or hinder billionaires that ultimately fund the work?

5. How good my research taste is. Are my ideas quack science? Has continuous interaction with sycophantic llms blinded me to developing appropriate critical faculties since I've been without regular technical supervision?
6. Is endlessly applying for insanely competitive fellowships even worth it? The expected value of completing an application is so low when one accounts for the finite-time they can be unemployed for. For example, Jason Tang (Bluedot technical AI safety) said applying for MATS is probably of a waste of time. When it began the acceptance rate was 30–50%. What do I do?
7. My strength relative to people on my physics cohort wasn't maths skill or dedication, it was agency to pursue questions no one else was thinking about. I could usually, in no small part due to communication and soft skills, persuade people these ideas weren't crazy. Given this combination of technical, soft-skills and communication, is this the most high impact, fulfilling thing I could be doing?
8. How stable is this as a career choice? How transferable are my skills and qualifications? Is there animosity toward AI safety orgs from academia and would transitioning be challenging if AI safety funding folds?
9. What my next steps are if I am not selected to work in this industry. There seems to be a paradox between the amount of money and the amount of outreach, and the selectiveness of the programs. Will I be discarded from this system and left empty-handed?

3.3 Skills required

Regarding the personal skills required, I have simply listed the skills I expect to be important for producing high-impact work in this kind of research.

- Research taste
- Convincing others that your idea is worth pursuing
- Marketing research effectively
- Maintaining integrity and not sacrificing research impact for career-ladder climbing
- Scout mindset
- Learning from smart people without slowing them down
- Offering value in unique ways that support more talented people
- Capacity for large-scale abstract thinking
- Resistance to “galaxy-brain” thinking
- Accepting that sometimes the highest-impact contribution is producing results that support governance work
- Being willing to sacrifice mathematical beauty in favour of ruthlessly high-impact research
- Being an LLM ‘power user’.
- Being resistant to LLM sycophancy.

4 MATS feedback

Thank you for requesting feedback on your MATS application. Below is a short summary of what stood out most in your application, where it fell short, and what we think would strengthen it.

Before you read: as noted on the opt-in form, this feedback was generated with AI assistance from our internal review notes, but reviews are imperfect. The points that resonate are probably worth reflecting on; the ones that feel off may say something about the limits of our process — or about something you could make more legible in a future application. On scope: we usually don't have insight into stream-level interviews or work tests — unless this feedback explicitly says otherwise, those were assessed by the streams directly and aren't reflected here. Below is our generated review of your application:

Your independent research execution was consistently valued. Your general-application answer about a problem you've spent 20+ hours on describes deriving a new in-degree distribution for finite growing networks and then, when no suitable goodness-of-fit test existed, building one — sole-authored, at "roughly 15 hours a week for 8 months", with your supervisor confirming both as extensions to the literature. The other recurring strength is care about when a statistical method's assumptions actually hold. In your Asymmetric Security answer about an evaluation you built, "The obvious but misleading move would have been to present that as uncertainty quantification full stop"; the account of why the diagonal Fisher breaks under attention and residual pathways, and the pivot to Monte Carlo dropout, read as judgment rather than a list of methods.

Where a future application would gain most is in making the risk-reduction argument explicit. Your research proposal for the Lee Sharkey stream engages closely with attribution-based parameter decomposition and connects it to information theory and geodesics, but doesn't set out what makes that direction safety-relevant — what harm becomes less likely if it works. The same pattern showed in your answer to the Peter Wildeford stream's governance-intervention question, where the biggest obstacle to systemic-risk evaluation of AI-mediated financial systems is "not yet having research output on the relevant mathematics" rather than the institutional and political barriers to regulators requiring it. The second point is the work tests themselves. Your research-taste work test came in near, but slightly under, expectations for this cohort. Your Applied AI assessment landed in the bottom half of the cohort. Both read as weaker than the application materials had suggested.

One concrete thing to do now: take a proposal you already have — the parameter-decomposition idea is the obvious candidate — and write the risk-reduction case as its own section, naming which failure it would make more detectable, who would act on the result, and what would have to hold for that to matter. Then ask someone working in safety research to attack that section. Second, LASR Labs runs a supervised team research cycle ending in a paper; applications are open until September 20, for a cohort running January to April. It would put you through a research cycle where the impact argument is required up front, which is the part your proposals currently leave to the reader. AISafety.com's events and training list tracks other openings if the timing doesn't work.

On this. I decided against applying for LASR by way of an expectation value calculation. I had a meeting the day after the deadline I prioritised. I am unsure if this was the correct move. I would welcome attacks to the *open abstract* detailed above permitting the criticism was constructive.

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